An Open Source Perovskite Solar Cell Curve Tracer

Walker S. Arce, *Member, IEEE*, Kelly Schutt, Joseph Luther, and Karen Stelling

***Abstract*—When developing systems for perovskite solar cells (PSCs), it is important to have a testing platform that closely resembles your deployed systems. While developing a 1 U CubeSat, the Big Red Sat-1 at the University of Nebraska-Lincoln, we found a lack of passive curve tracing instrumentation, which we needed for our mission. To that end, we developed a benchtop passive curve tracer that leverages the Ossila Solar Simulator as an AM1.5 light source. This device utilizes a ladder of switched resistors and a precision voltage and current measurement to achieve the curve tracing. It also integrates a common-anode multiplexer to switch through multipixel PSCs. Finally, the solar cell holder is separate from the main measurement system, so various substrates sizes and pixel counts can be supported. This system is tested using various PSCs. It has been open sourced to enable reuse in future systems that may benefit from the lower power consumption and flight heritage.**

***Index Terms*—Solar energy, perovskites, open source hardware, open source software.**

I. Introduction

PEROVSKITE solar cells (PSCs) have demonstrated the fastest efficiency scaling of any solar power device [1]. Both terrestrial and space applications benefit from the low manufacturing cost, recyclability, and high power density of PSCs [2], [3]. In addition, the absorber material can be transported to off-world surfaces, where 12 kg of material can fabricate a 12-MW solar array [3]. To assess the performance of PSCs in real-world environments, testing apparatuses need to be developed and deployed to measure key parameters. Many key parameters of PSCs can be captured by curve tracing.

A curve tracer measures the development of voltage and current coming from, or going into, a device while varying the load conditions, the resulting curve being termed an *I–V* curve [4]. These two variables are plotted with voltage on the *x*-axis and current on the *y*-axis, yielding a variety of common shapes. The modulation of load conditions is a well known problem in electrical engineering and there are many solutions for curve tracers, including a capacitive load, an electronic load, and a resistive load [4]. All topologies utilize a voltage measurement block across the load and a current measurement block in line with the solar panel. The capacitive load has a capacitor attached to the solar panel and the charge cycle of the capacitor allows the curve to be captured [5]. The electronic load method uses a MOSFET in the triode region to create a variable resistance, so a modulated gate signal proportionally modulates the load [6]. Finally, the variable resistance method is very similar to the electronic load method, but discrete resistors are used along with switches to connect and disconnect those resistors from the load [7].

Curve tracing during real-world operation of solar cells can indicate their health and monitor for failure conditions [4]. Any degradation or change in performance of the solar cells will produce a signature in the *I–V* curve [4], [8], which when paired with other environmental conditions can indicate how failures happen [9], [10]. When transitioning PSCs from the laboratory to the field, various experiments will be performed in a variety of environments. Some environments, like space, introduce constraints on real estate, power budget, and complexity. To facilitate preparation for practical testing in space environments, we present a low-cost, open-source, benchtop curve tracer for PSCs. This system was designed for testing PSCs in preparation for the Big Red Sat-1 CubeSat mission [8], which successfully operated in orbit for 105 days while extracting JV curves from three PSC samples using the presented circuits [11].

The presented design is also applicable to other solar cell architectures, provided the voltage and current characteristics of the solar cell are similar to PSCs. Alternatively, the design can be modified to accommodate higher voltage and current solar cells through simple modifications.

II. Methods

A. Curve Tracer

The basic building blocks of a curve tracer consists of a voltage measurement circuit, current measurement circuit, and a load modulating circuit [4]. The first two circuits are straightforward to implement. Considering the PSC pixels have an open-circuit voltage around 1 V, the voltage measurement can be accomplished using a voltage-follower operational amplifier circuit. Similarly, current measurements can be performed using a series shunt resistor and measuring the burden voltage to calculate the loop current.

To vary the load, the variable resistance method is implemented. This method provides the means to control the precision of the *I–V* points and can simplify the system design and control. The main design consideration is the selection of a switch to select, or bypass, the resistor. If eight resistors and switches are chained together, then 8-b of control are available, meaning that 256 *I–V* points are possible. Fig. 1 shows the two main load configurations.

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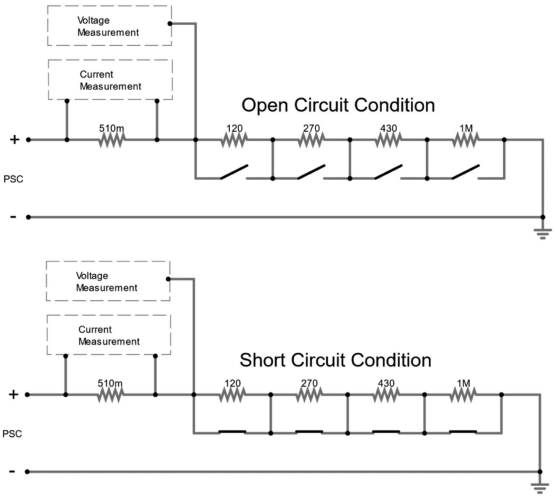


Fig. 1. Top shows the resistor and switch configuration for the open circuit condition and bottom is the resistor and switch configuration for the short-circuit condition. The resistor values are from the electronics used in the Big Red Sat-1 mission [8].

The open-circuit condition for a PSC generates maximum voltage and minimum current, while the short-circuit condition generates minimum voltage and maximum current.

B. Common-Anode Multiplexer

Generally, a curve tracer for a single solar cell is not as practical as one that can curve trace multiple in a single system and isolate the individual cells. This can be accomplished in field systems by using a double pull double throw (DPDT) relay that has the normally closed node connected to the solar charge controller and the normally open node connected to the curve tracer [12], [13]. This works well for terrestrial systems, where an atmosphere of pressure can be roughly assumed. This also works well for solar cells that have a single positive and negative lead that is strung together in a system with a common cathode, so removing the positive connection is straightforward. For laboratory scale PSCs, they are regularly fabricated with multiple power generating "pixels" with a common-anode [8] due to its p-i-n construction [14]. It is common for these PSCs to have six pixels, as shown in Fig. 2, and a practical curve tracer needs to multiplex the ground connections from each pixel.

For p-i-n PSCs, the ground node for each pixel has its connection controlled. Fig. 3 shows the two connection choices. Since most PSCs have six pixels, the multiplexer would need six switches to control the measurement of the entire PSC sample.

PSCs have the potential to degrade from different load conditions and exposure to light causes the generation of iodide vacancies that increases ionic mobility, leading to degradation mechanisms and the alteration of device behavior [15]. Because of this effect, there should be an electrical connection for removal of charged iodide vacancies. There are three options: 1) open circuit; 2) short circuit; or 3) maximum power point. The impact on lifespan of these three conditions was characterized in [16], showing that maximum power point is the ideal bias point. Each condition will be tested.

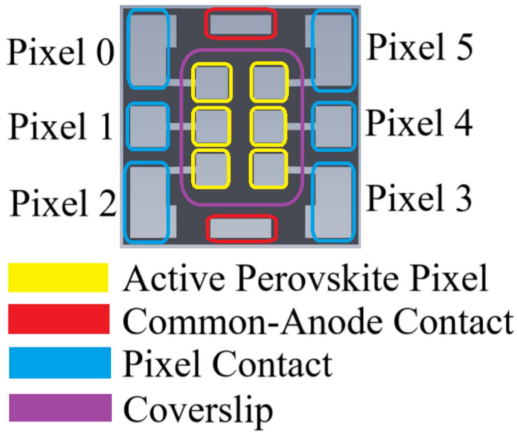


Fig. 2. Common configuration for laboratory PSCs, showing six pixels, the metallic contacts, a common-anode contact, and a coverslip for protection from environmental conditions.

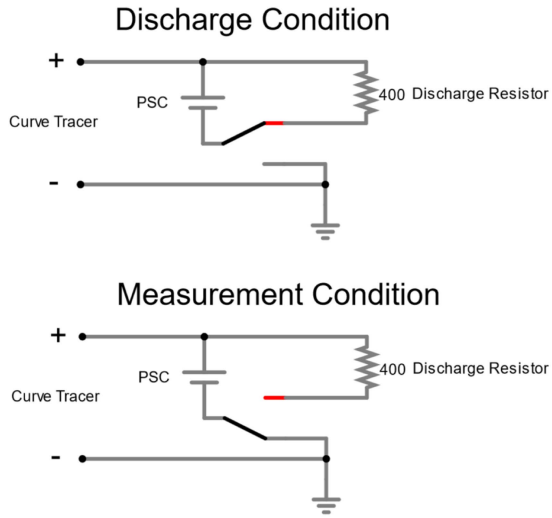


Fig. 3. Top shows the discharge condition of a PSC using a DPDT relay and bottom shows the measurement condition of the same.

C. Light Source

To test the power generating capabilities of a PSC, a calibrated light source needs to be used. There are many options on the market, but the smallest and most easily integrated system is the Ossila LED Solar Simulator. This solar simulator produces a calibrated AM1.5 light that meets the IEC 60904-9:2020 standard using LEDs. The light meets class AAA in a 15-mm diameter area and class ABA in a 25-mm diameter area. An optical breadboard acts as the mechanical base for the device, providing ample space for hard mounting of electronics. In addition, this solar simulator provides a serial control interface and shutter capabilities.

D. Software

The Python programming language [17] will be used to command the curve tracer and receive *I–V* points from the measurement circuits. A simple graphical user interface will be designed that allows user control and real-time visualization of *I–V* curves. An MSP430 microcontroller from Texas Instruments

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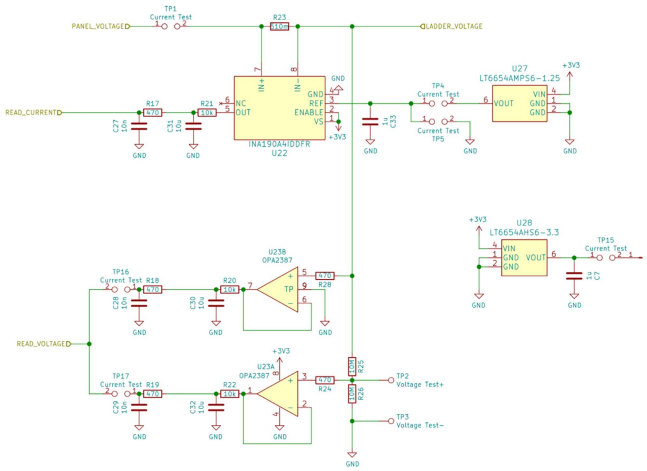


Fig. 4. Schematic of the curve tracer showing the voltage and current measurement circuitry.

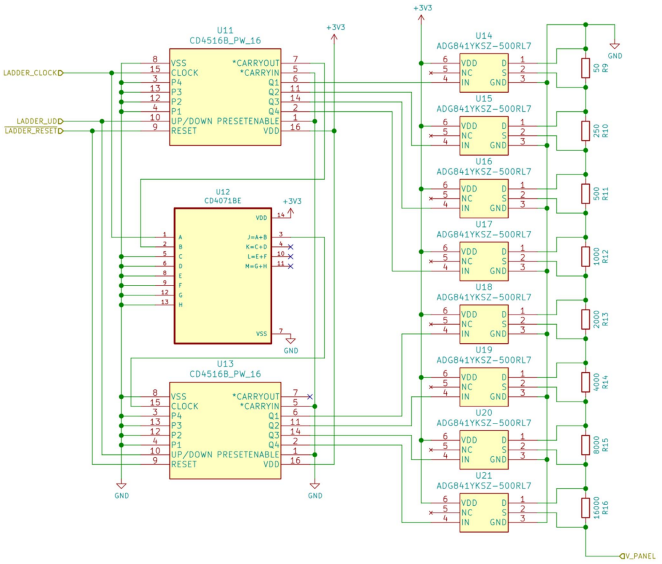


Fig. 5. Schematic of the resistor ladder, showing the counter circuits and SPST analog switches.

will be integrated to control the curve tracer and multiplexer, which will be programmed in C++. A serial interface command set will be implemented that allows hardware control.

III. Results

A. Hardware

The electronics were designed using KiCad 5.1.10 and were assembled by hand. The MSP-EXP430FR5969 LaunchPad development kit was integrated for the embedded control, which allowed easy adaptation to other LaunchPad development kits. The main circuits are shown in Fig. 6 and are flight tested [11].

Fig. 4 shows the curve tracer. The voltage measurement is performed using an OPA2387 operational amplifier, providing an ultra-high precision, zero-drift, low-input bias current interface, and characterized from −40 °C to 125 °C. Two channels provide either a voltage follower or 0.5× amplification. The current measurement is performed using an INA190A4 current shunt amplifier, providing 200× amplification of the shunt voltage,

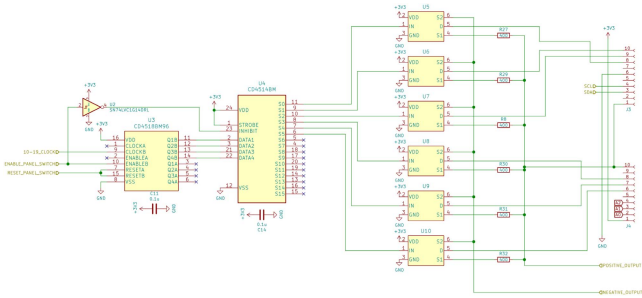


Fig. 6. Schematic of the multiplexer, showing the counter circuit and the SPDT analog switches.



Fig. 7. Curve tracer in use with the Ossila Solar Simulator, with marked dimensions for scale.

zero-drift, and characterized from −40 °C to 125 °C. For the current sensor, a precision reference generator of 1.25 V can be chosen using a jumper. An additional 3V3 precision reference generator is provided for the microcontroller, which is connected in using a jumper.

Fig. 5 shows the resistor ladder for varying the load conditions of the PSC. A cascaded 8-b binary up/down counter is implemented using a CD4516, with a CD4071 to handle overflow ripple. The SPST switches control the connection or bypassing of the selected resistor using the ADG841, providing a 370-mΩ conduction channel, which is significantly lower than any resistor on the ladder. The selected resistors are 50 Ω, 250 Ω, 500 Ω, 1 kΩ, 2 kΩ, 4 kΩ, 8 kΩ, and 16 kΩ, from least significant bit (LSB) to most significant bit (MSB). It is important to have each successive resistor to be twice as large as the preceding one, to allow a linear increase as the ladder is incremented.

Fig. 6 shows the common-anode multiplexer circuit, which selects the pixel of the PSC to be measured. A BCD-to-decimal counter is implemented using CD4518 cascaded with a CD4514. The SPDT switches are implemented using ADG839, with 500-mΩ conduction channels, where the normally connected channel (S1) is connected to the discharge path and the normally open channel (S2) is connected to the negative lead of the curve tracer.

The final assembled and in-use curve tracer is shown in Fig. 7, with the Ossila Solar Simulator illuminating a PSC under test. The USB port on the LaunchPad development kit is used to command the instrument.

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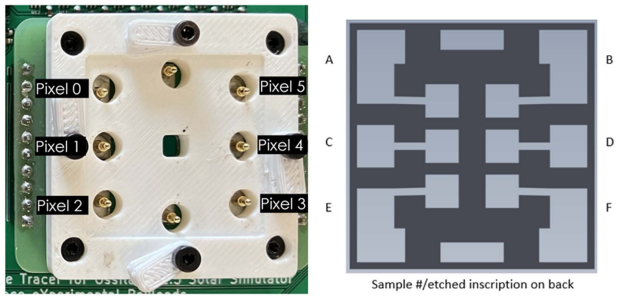


Fig. 8. Developed sample holder on the right, showing the gold pogo pins, 10 pin contacts, and pixel pinout. On the left, the corresponding pixel names for the National Laboratory of the Rockies (NLR) samples used for comparison.

This setup is ideal for benchtop experiments with solar cells in an inert atmosphere glove box. While this final circuit board would (most likely) not be incorporated into a CubeSat, the actual circuits can be fit into a compressed area. As an example, Big Red Sat-1 incorporated similar, and additional, circuitry to support the mission and the flight computer for that mission was 3.19-by-2.95 in [8]. This flight computer is also open-source as an example of how this system could be utilized for space missions.

B. Sample Holder

The sample holder was designed to allow quick switching of PSCs by integrating the sample holder onto two 10-pin headers. A 3-D printed frame is screwed into the circuit board that has rotating fingers that hold the PSC onto the frame. On the circuit board are gold-plated pogo pins that provide electrical contact to the PSC, which the rotating fingers apply pressure to compress the pogo pins with the PSC. To control the power generating area of the PSC, a light mask is layered on top of the PSC. The PSC electrical connections are routed out of the circuit board through the 10-pin headers. The sample holders for the PSC, shown in Fig. 8, were designed using Fusion 360 and printed on a Sovol SV06 3-D printer using white polylactic acid (PLA) filament.

C. Software

To communicate with the Ossila Solar Simulator, a Python library was developed called pyOssila which implements the full command specification for use in scripts [18]. Fig. 9 is the user interface developed using tkinter, which is built into Python. When running a test the following information can be entered: the ID of the PSC, the power generating area of the PSC, and the irradiance being tested. In addition, the user can indicate that the curves should be saved using the "Save Curves?" checkbox. Once those parameters are setup, the "Trace Curve" button can be pressed to initiate the characterization. As curves are captured from the hardware, the curves are displayed in the plot on the lower half of the display.

For the purposes of comparison, the current density was calculated instead of utilizing only current, yielding a *J–V* curve, instead of an *I–V* curve. The only difference is that the current at each point is divided by the power generating area of the PSC under test.

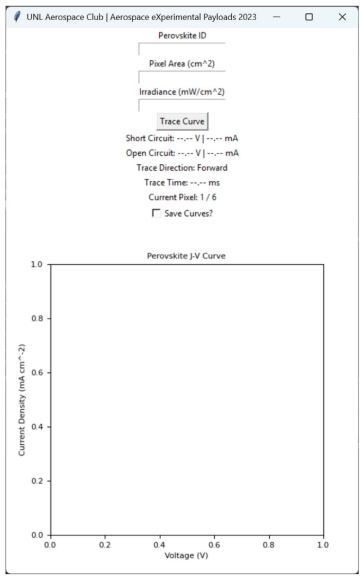


Fig. 9. Curve tracer in use with the Ossila solar simulator illuminating the sample.

TABLE I
BYTE CODE COMMAND SET FOR THE CURVE TRACER

Byte Code	Function	Utility
0x43	Trace curve	Completes a curve trace using the current selected pixel and trace direction, streams 512 bytes of raw current and voltage data.
0x44	Get current pixel	Returns the currently selected pixel on the multiplexer.
0x45	Increment multiplexer	Steps the multiplexer counter to the next multiplexer, automatically rolls over to first pixel.
0x46	Toggle trace direction	Inverts the trace direction between forward and reverse, depending on current setting.

The firmware for the MSP430 was written in C++ using the Energia Integrated Development Environment and implements the analog-to-digital conversion, the control of the curve tracer, and control of the multiplexer. A simple byte-code command structure was implemented, consisting of four commands which are explained in Table I.

D. Curve Tracing

The developed system was used extensively in the Big Red Sat-1 project for characterizing samples provided by the National Laboratory of the Rockies. Different discharge loading conditions were tested, which presented an issue for the multiplexer. The resistive load curve tracer varied from around 2 Ω to 1 MΩ. When the discharge resistor was set at any point above 2 Ω, there would be leakage from the other pixels, causing a surge in the *J–V* curve, as shown in Fig. 10 with PSC X032.

This left only the second best choice for the discharge resistor, which was short circuit. Once the discharge path has a lower resistance than the short-circuit condition of the curve tracer, then no leakage is observed. So, this resistor was fixed to 0 Ω. This solves the interpixel interference caused by the multiplexer. After resolving this question, the system was prepared for characterizing samples, as shown in Fig. 7.

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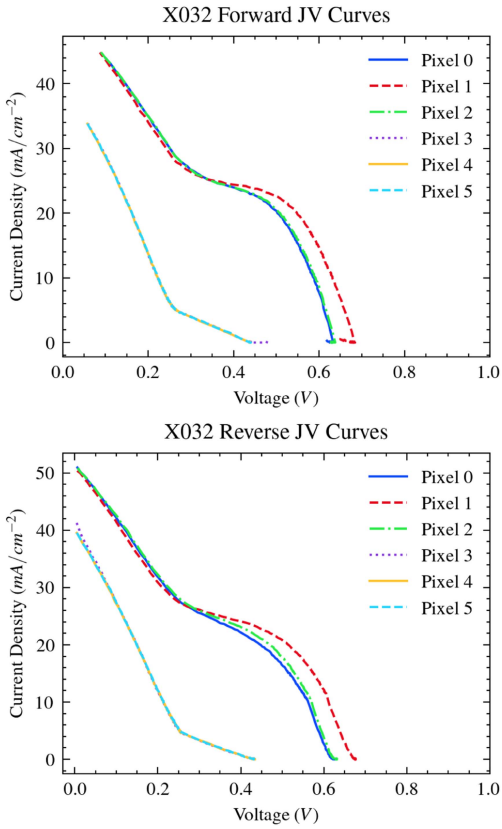


Fig. 10. J–V curve would surge from a discharge resistor that was set in the active curve tracing range. This figure shows the discharge resistor set to 500 Ω, and when the curve tracer went below this value, the other pixels would leak into the measurement node and cause a surge in the curve. The top figure is the forward trace condition and the bottom figure is the reverse trace condition.

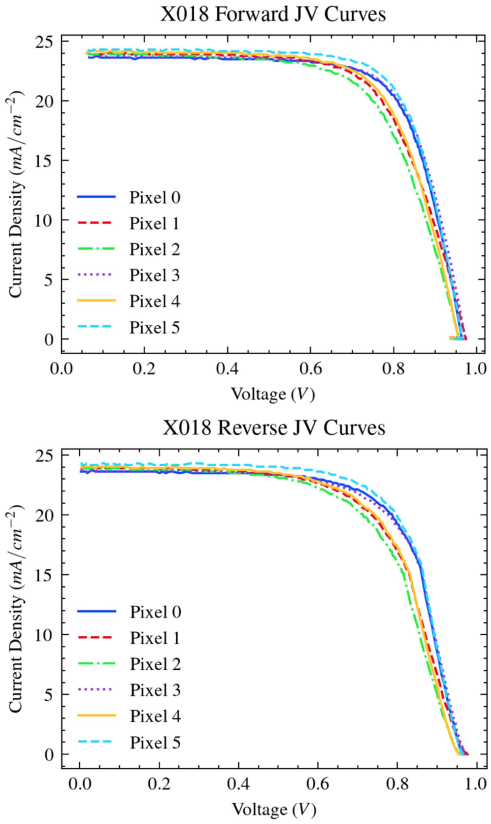


Fig. 11. Curves traced on the X018 sample, top is the forward condition and bottom is the reverse condition.

NLR provided three rounds of PSCs for testing, which were all characterized using this hardware, the data from this testing can be seen here.[1] Fig. 11 shows the data for sample X018, which was launched on Big Red Sat-1 [8] and was the champion PSC that maintained T80 while in low Earth orbit, as discussed in [11]. The curves show all six pixels operating with a short-circuit current density near 25 mA cm^{-2}, corresponding to a percent efficiency between 15% and 18%, depending on the pixel. Some hysteresis can be observed between the forward and reverse trace condition, with a slightly more rounded knee on the reverse condition.

E. Keithley Comparison

All of the PSC samples used to develop the presented curve tracer were first characterized by NLR using their Keithley 2400 series SourceMeter and Sunbrick G2V Class AAA LED light source. For the presented PSC, X018, NLR characterized the sample on 8 September 2023 and UNL characterized the same sample on 16 October 2023. The extracted parameters from both systems are presented in Table II.

These results were taken 38 days apart with the sample being shipped between Colorado and Nebraska. While some degradation may have occurred, it is also possible that the presented curve tracer has some inaccuracy that could be calibrated out through the use of a calibrated voltage and current source. The calibrated voltage and current source should be supplied to all six channels simultaneously. This would provide correction factors that would align the measurements with known quantities. Sources of error in the measurements are the sense resistor (which induce a burden voltage error), gain error, and interpixel interference. The first two can be addressed through standard calibration procedures and the third can only be addressed through the discharge resistor on the normally closed path of the multiplexer. Without calibration, the extracted parameters may be slightly high or low compared with ground truth, but the electrical circuits are high-precision devices with some compensation already built in. In addition, the resistors selected were chosen to have automotive quality ratings, low temperature coefficients (± 10 ppm/∘C), and low variability (±0.05%) to ensure consistent operation between temperature extremes and production batches.

TABLE II
COMPARISON BETWEEN PRESENTED CURVE TRACER (UNL COLUMNS) AND A LABORATORY KEITHLEY SYSTEM (NLR COLUMNS)

	$V_{OC}(V)$		J_{SC}(mA/cm^{-2})		FF (%)		Efficiency (%)	
	NLR	UNL	NLR	UNL	NLR	UNL	NLR	UNL
Pixel 0	1.09	0.97	22.20	23.80	67.50%	69.40%	16.30%	15.98%
Pixel 1	0.00	0.98	0.00	23.90	0.00%	63.70%	0.00%	14.87%
Pixel 2	1.10	0.96	22.60	23.90	74.70%	62.40%	18.50%	14.33%
Pixel 3	1.10	0.98	22.50	23.80	74.30%	67.40%	18.40%	15.62%
Pixel 4	1.09	0.96	22.50	24.00	68.40%	64.80%	16.70%	14.93%
Pixel 5	1.09	0.97	22.20	24.30	73.30%	69.70%	17.70%	16.37%

[1][Online]. Available: https://github.com/Big-Red-Sat/Perovskite-Curve-Tracer/tree/main/software/curves on GitHub

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One thing to notice is that Pixel 1 had no parameters extracted by NLR. X018 had a SiO coverslip epoxied to it to protect the perovskite active layer from the environment [2], but the epoxy can also inadvertently cover up contacts. The benefit of the developed sample holder is that the pogo pins can have their position adjusted because of the large copper pad provided, so getting contact in regions closer to the edge of the sample are possible. This allowed Pixel 1 to have its parameters extracted by the UNL team.

IV. Discussion

While the curve tracer was successful in completing our mission and performed well, there is always room for improvement. The multiplexing circuitry was chosen to be compatible with space conditions, where semiconductors are less prone to environmental damage due to low pressure and temperature fluctuations. It was also deemed less risky to the overall mission. An alternative would be to follow the approach of [12] and [13] and use electromechanical relays.

For space environments, this could be risky or require hermetically sealed relays. The risk comes from the evacuation of the air in relay from the vacuum of space, which can lead to arc-at-break damage [19]. Teverovsky [19] demonstrated this failure mode with switching high power antenna feedlines, so it is possible that the lower power of the PSCs would not have the same issues. For terrestrial systems, or near-space conditions, this would be less likely to be a problem. This would also make the multiplexing consume more power, but would remove the issue with the leakage that was demonstrated in Fig. 10. This could improve the lifetime of the PSCs when in storage, as the normally connected node would always be engaged, as opposed to the OFF-state of the semiconductor SPDT switch.

While the curve tracer was designed to enhance the testing procedures for Big Red Sat-1, we recognize that this test setup may be useful to other teams. For that reason, the project has been made open source.[2] The hardware is licensed under a CERN Open Hardware Licence Version 2 - Strongly Reciprocal, the software is licensed under an MIT license, and the documentation is licensed under a Creative Commons Attribution-ShareAlike 4.0 International Public License. The described circuits and components were validated by the Big Red Sat-1 mission [8], [11], which also made its designs open-source, including its flight computer. Future teams may consider utilizing the resistor ladder curve tracing approach to reduce the power requirements of your mission. In addition, the multiplexer has proven effective in limiting the interpixel interference. The overall circuitry can be compressed into a small region, which allows a scale up of the system to multiple solar cell samples.

These circuits are also compatible with curve tracing of other solar cell architectures. The only limitation present is the voltage and current measurements required. Depending on the voltage of your solar cells, this may require the use of a different multiplexer. If your short-circuit current is higher than a PSC, then a different sense resistor and resistor ladder values would be required.

[2][Online]. Available: https://github.com/Big-Red-Sat/Perovskite-Curve-TraceronGitHub

V. Conclusion

For future missions that investigate PSC performance in space missions, the reported curve tracer provides a low-cost and easily understandable instrument for characterizing PSCs. The system is fully integrated, with an AM1.5 light source, hardware unit, and software packages for usage, which allows quick turnaround on laboratory usage. The open-source hardware allows integration into flight computers and allows modifications for new parts as improvement of electronics continues over time. The hardware designs are licensed under CERN Open Hardware License Version 2, the software is licensed under an MIT license and the documentation is licensed under a Creative Commons Attribution-ShareAlike 4.0 International Public License.

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